SkyLedger

OPEN AGENT HACKATHON 2026 TRACK 1 · SOLVING FRAGMENTED INTELLIGENCE

Satellites see the leaks.
SkyLedger checks the reports.

An AI agent team that checks satellite-observed methane against what oil and gas facilities report to the US EPA, and who owns them. Every finding is ranged, traced, and challenged by a Skeptic before an analyst signs it.

- Zetaris
- Meterless Nightshift
- Meterless World Model
- NVIDIA Nemotron
- Cursor

github.com/skyledger-org/skyledger

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01 THE PROBLEM

Three sources that were never designed to meet.

SKY · CARBON MAPPER

kg / hour

A plume is a point and a rate

Seen from aircraft and satellites (Tanager, EMIT, AVIRIS-NG, GAO) on one morning. For example **207 kg/h** on 2026-09-05.

GROUND · US EPA GHGRP

t CO_2e / yr

A report is a facility and a year

Self-reported once a year, sometimes for a whole basin. Tiger Plant reported **16 t CH_4** for RY2023.

LEDGER · OWNERSHIP

free text

An owner is a string

`"TARGA RESOURCES (73%); PIONEER NATURAL RESOURCES USA INC (27%)"`

Spelled many ways, no IDs.

18 of 60

Permian plumes where matching to the nearest facility blames a plant **more than 1 km from where the plume starts**. No shared keys, units or clock, so nobody checks one against the other.

02 WHO IT'S FOR, AND WHY NOW

For the people who must judge whether a reported number is plausible.

ESG and credit analysts

Signing off on issuers' emissions disclosures, and pricing transition risk.

Regulators and state agencies

Deciding which facilities to inspect first, with evidence that survives challenge.

Journalists and NGOs

Who need traceable evidence, not anecdotes, before naming a company.

WHY NOW

10 Nov 2026

California **SB 253**: first Scope 1 and 2 reports are due, with no assurance requirement yet.

RY2025

EPA has **delayed GHGRP reporting** and proposed ending most of the program. Independent checks matter more.

Public & frequent

Satellite methane data is now open (Carbon Mapper, Tanager).

03 THE SOLUTION

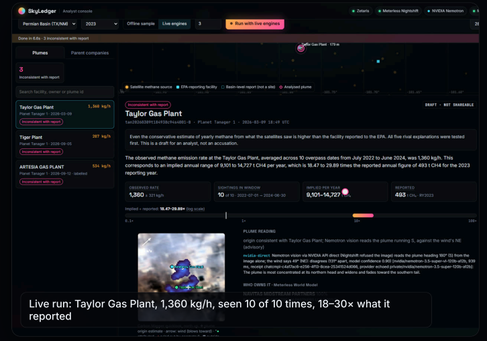
Every plume becomes a traced, ranged, signed-off dossier.

1 · Find
Federated geo-join of plumes to reporting facilities, queried in place through Zetaris.

2 · Attribute
Wind and plume geometry pick the facility at the plume's head. Nemotron reads the image as a check.

3 · Reconcile
kg/h becomes tonnes per year, always a range, inside the reporting window. Owners resolve to parents.

4 · Challenge and sign
A Skeptic rules out five innocent explanations first. Nothing is shareable until an analyst signs.



04 AGENT WORKFLOW

Eight agents, one shared state, and a Skeptic with veto power.



Not a linear chain

The Skeptic sends work back: widen the search (GeoLinker), look again (PlumeReader), or stop annualising (Reconciler). Loops are bounded.

Numbers come from code

Distances, unit conversion and ratios come from SQL and code. Models read images and word findings; a guard rejects any number the code didn't compute.

Traceable

Every step goes to a hash-chained JSONL trace that streams live to the console. Editing any line breaks the chain.

05 THE SKEPTIC

Before a word is written, five rival explanations must fall.

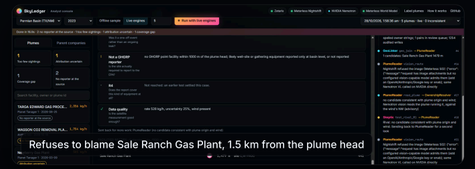
R1 **Wrong facility**
Another site fits the plume head and wind, or the nearest reporter is over 1 km away.

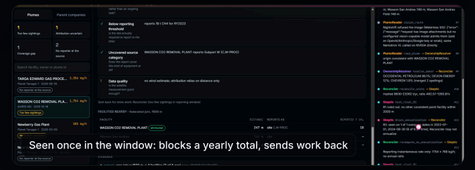
R2 **One-off release**
Fewer than 2 sightings in the reporting window blocks any yearly total.

R3 **Not required to report**
No subparts, no CH_4 filed, or no reporter at the plume head.

R4 **Uncovered equipment**
Only basin-level reports nearby, or no Subpart W coverage.

R5 **Bad measurement**
No rate, uncertainty above the limit, or no wind.





SkyLedger

Skeptic

06 WHAT IT FINDS · RUN EXAMPLE-1, PERMIAN BASIN (TX/NM), GHGRP RY2023

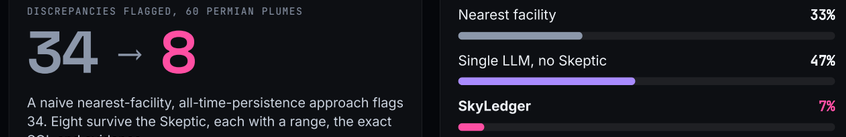
Fewer flags. Better ones.

DISCREPANCIES FLAGGED, 60 PERMIAN PLUMES

34 → 8

A naive nearest-facility, all-time-persistence approach flags 34. Eight survive the Skeptic, each with a range, the exact SQL and evidence.

WRONG FACILITY BLAMED, 15 HAND-LABELLED PLUMES



A small sanity check labelled by the team, not expert ground truth. Details in eval/results.md.

3,413
satellite sources in the scan area

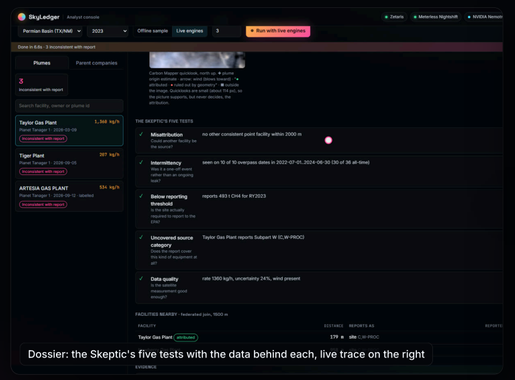
31
plumes with no reporting facility at the head

10
too few sightings to estimate a year

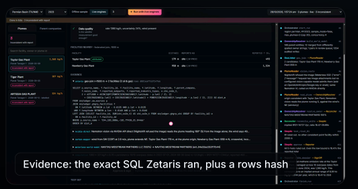
0
false discrepancy flags on the labelled set

07 THE PRODUCT

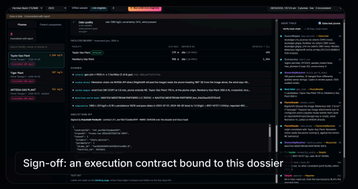
An analyst console, not a chatbot.



Dossier: the Skeptic's five tests with the data behind each, live trace on the right



Evidence: the exact SQL Zetaris ran, plus a rows hash

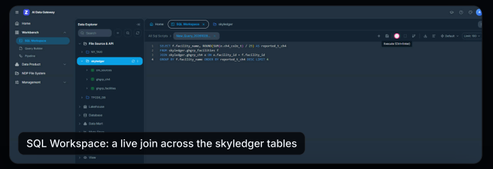


Sign-off: an execution contract bound to this dossier

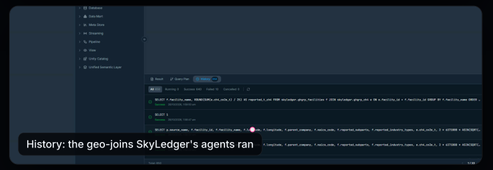
08 SPONSOR TECHNOLOGY · ZETARIS

Zetaris is the data layer: joined in place, never copied.

- **Three public sources** registered as Zetaris Lightning tables: Carbon Mapper sources, GHGRP facilities, GHGRP CH_4 by subpart.
- **One federated geo-join per plume** through the REST SQL API: bounding box plus haversine distance, where the data lives.
- **SourceMapper** discovers the reachable tables and row counts at the start of every run.
- **Evidence:** the exact SQL and a rows hash in every dossier, so an analyst can rerun it.
- **Verified:** Zetaris equals the offline cache for **60 / 60** plumes, and the agents' queries appear in Zetaris's own history.



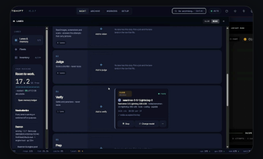
SQL Workspace: a live join across the skyledger tables



History: the geo-joins SkyLedger's agents ran

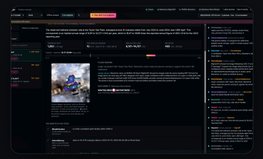
09 SPONSOR TECHNOLOGY · METERLESS AND NVIDIA

Meterless runs the models and the ownership graph. Nemotron reads and writes.



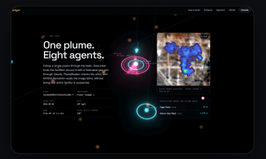
Meterless Nightshift

The primary AI provider, built to Meterless's provider contract (headers, discovery, token handling, Auto Detect). The NVIDIA key lives inside Nightshift, not in our app. Contract checks pass.



Meterless World Model

The facility → owner → parent graph, with provenance on every write, merged spellings and a review queue. Meterless's conformance suite: **8 / 8**. Sign-off uses Scout Intent's contract format.



NVIDIA Nemotron

3.5 Lightning via Nightshift writes each dossier under a number guard. **3.5 Super VL** reads the plume image blind: 38% agreement with the wind (chance ≈ 33%), so it's advisory. VL fallback to NVIDIA's API is disclosed and logged.

10 TECHNICAL STACK, AND WHAT WE MEASURED

Small, reproducible, and honest about what's live.

STACK

- **TypeScript on Node 22**, run with tsx. No build step, no GPU.
- **HTTP API:** POST /run, live server-sent-event streaming, signed sign-off.
- **Front end:** vanilla JS, three.js, GSAP and Lenis, served locally.
- **Config, not constants:** every threshold in config/skyledger.json, regions in config/regions/.
- **Sample mode and Docker:** runs from a clean clone with no keys and no network.

MEASURED, NOT CLAIMED

60/60	Zetaris geo-join equals the offline cache
8/8	Meterless World Model conformance suite
38%	Nemotron vision vs the wind (chance ≈ 33%), so advisory
7%	wrong facility blamed on hand labels (nearest facility: 33%)
✓	clean-clone Docker run, no .env, no network

11 FUTURE ENHANCEMENTS

From one basin to a monitoring service.

Coverage

Every US basin and reporting year, scheduled refresh from Carbon Mapper and EPA, SEC CIK links for parent companies.

Accuracy

A larger expert-labelled test set, vision on larger scenes as a data-quality check, and learning attribution rules from analyst decisions.

Product

A hosted service with logins and audit trails, alerts when a facility's gap persists, and signed dossiers exported for disclosure reviews.

SkyLedger

Check what satellites saw against what was reported.

`github.com/skyledger-org/skyledger`
`runs offline: npm ci && npm start`

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